

stats101a_hw7

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.2      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

(1)

```
df <- data.frame(read.csv('./AdRevenue.csv'))

str(df)
```

```
## 'data.frame':   70 obs. of  4 variables:
## $ Magazine      : chr  "People" "Better Homes and Gardens" "Time" "Parade (1)" ...
## $ PARENT.COMPANY..SUBSIDIARY: chr  "Time Warner, (Time Inc.)" "Meredith Corp." "Time Warner, (Time Inc.)" ...
## $ AdRevenue      : num  233 397 286 877 304 ...
## $ Circulation     : num  3.75 7.64 4.07 32.7 3.21 ...
```

Part A

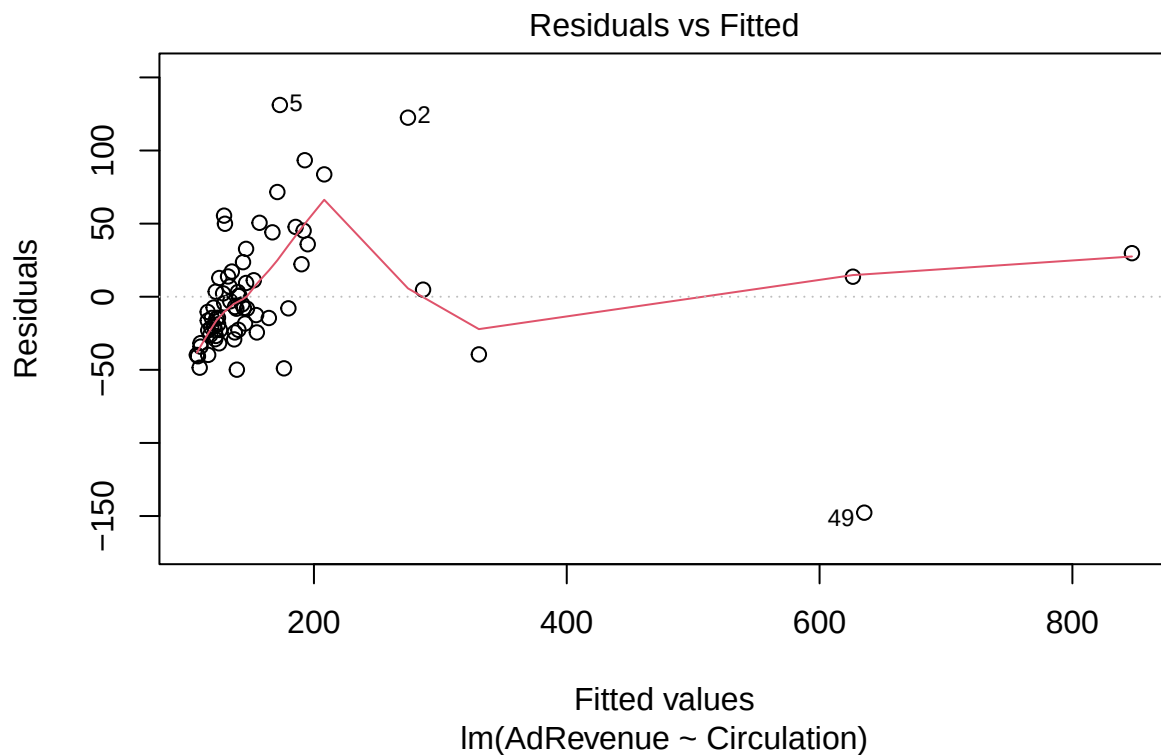
(a)

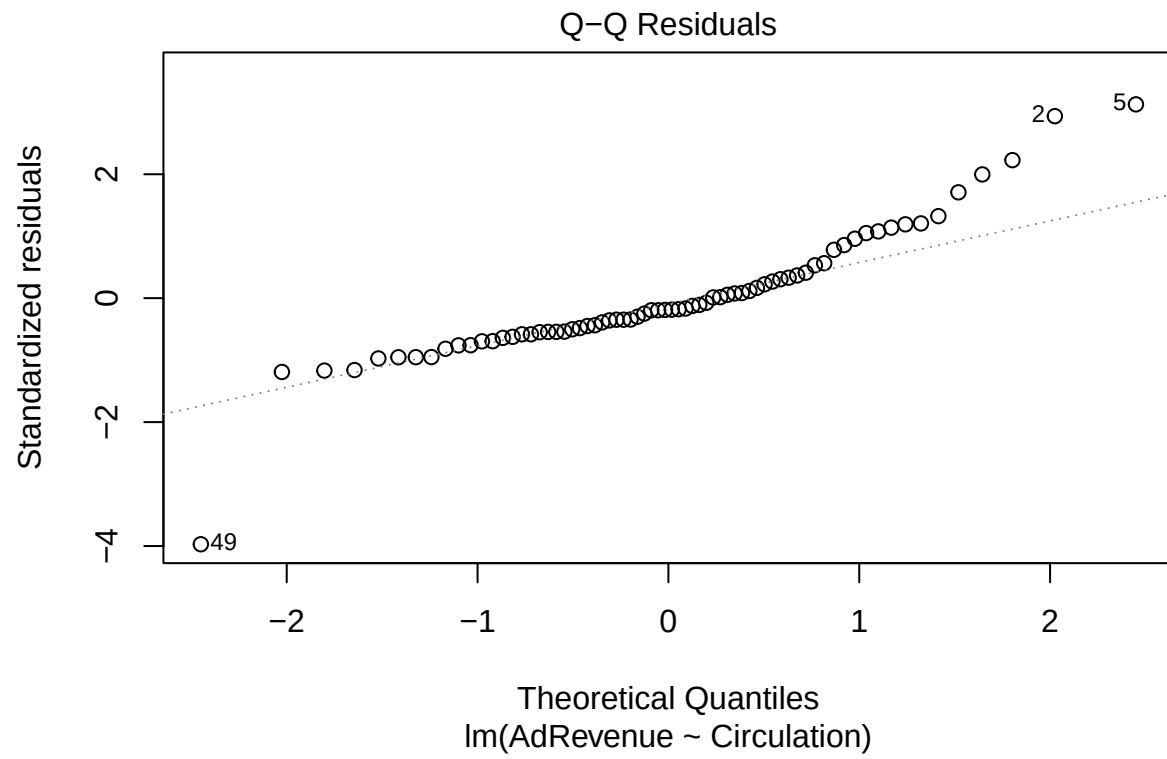
```
model <- lm(AdRevenue ~ Circulation, df)

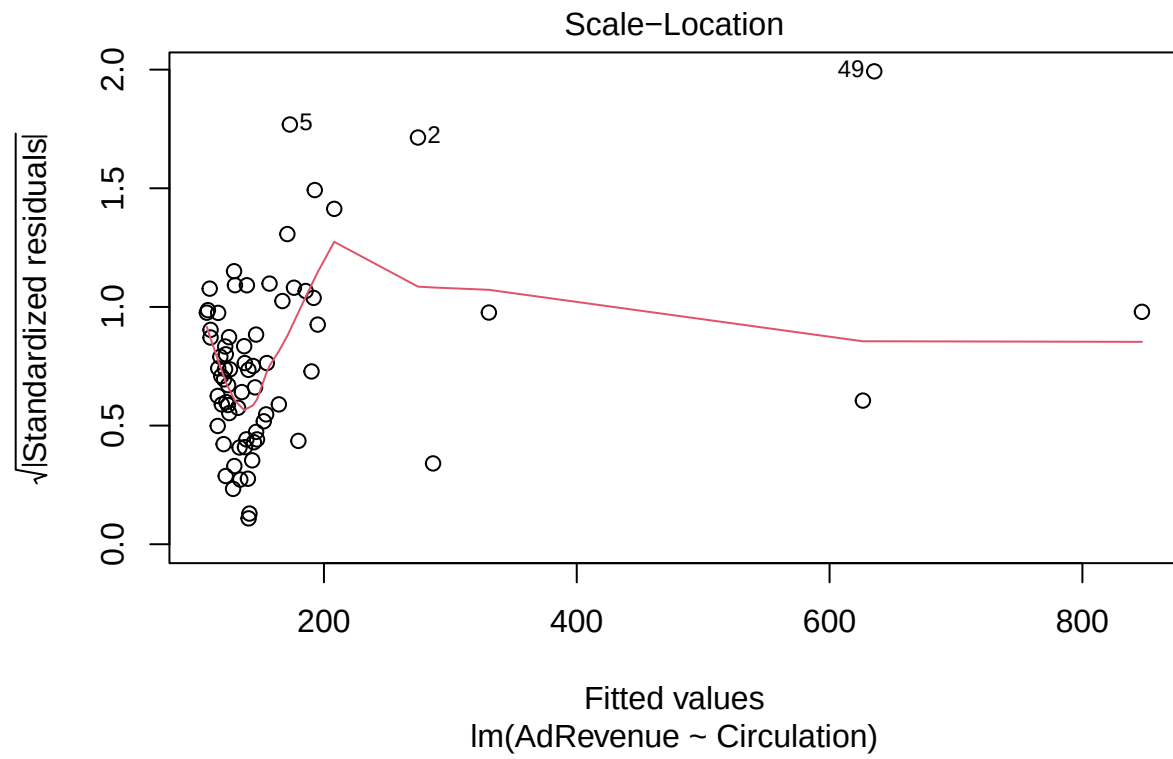
summary(model)
```

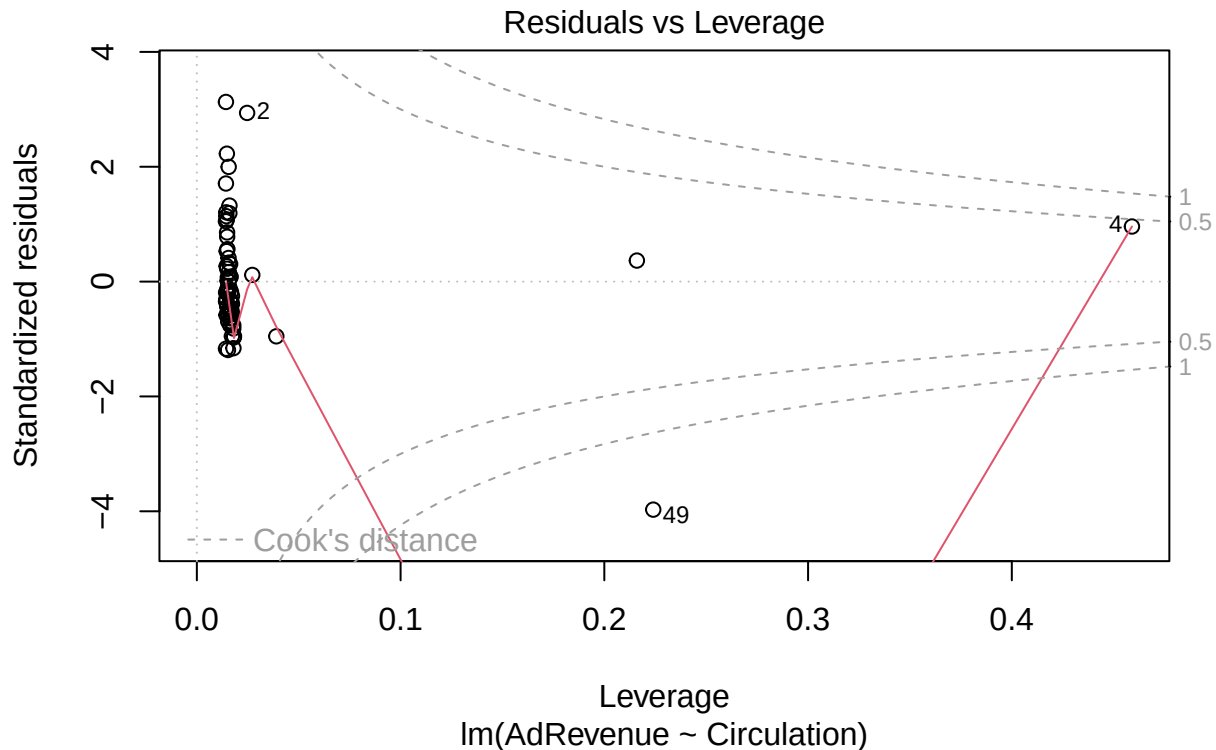
```
##
## Call:
## lm(formula = AdRevenue ~ Circulation, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -147.694  -22.939   -7.845   13.810  131.130
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   99.8095     5.8547   17.05  <2e-16 ***
## Circulation   22.8534     0.9518   24.01  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 42.22 on 68 degrees of freedom
## Multiple R-squared:  0.8945, Adjusted R-squared:  0.8929
## F-statistic: 576.5 on 1 and 68 DF,  p-value: < 2.2e-16
```

```
plot(model)
```







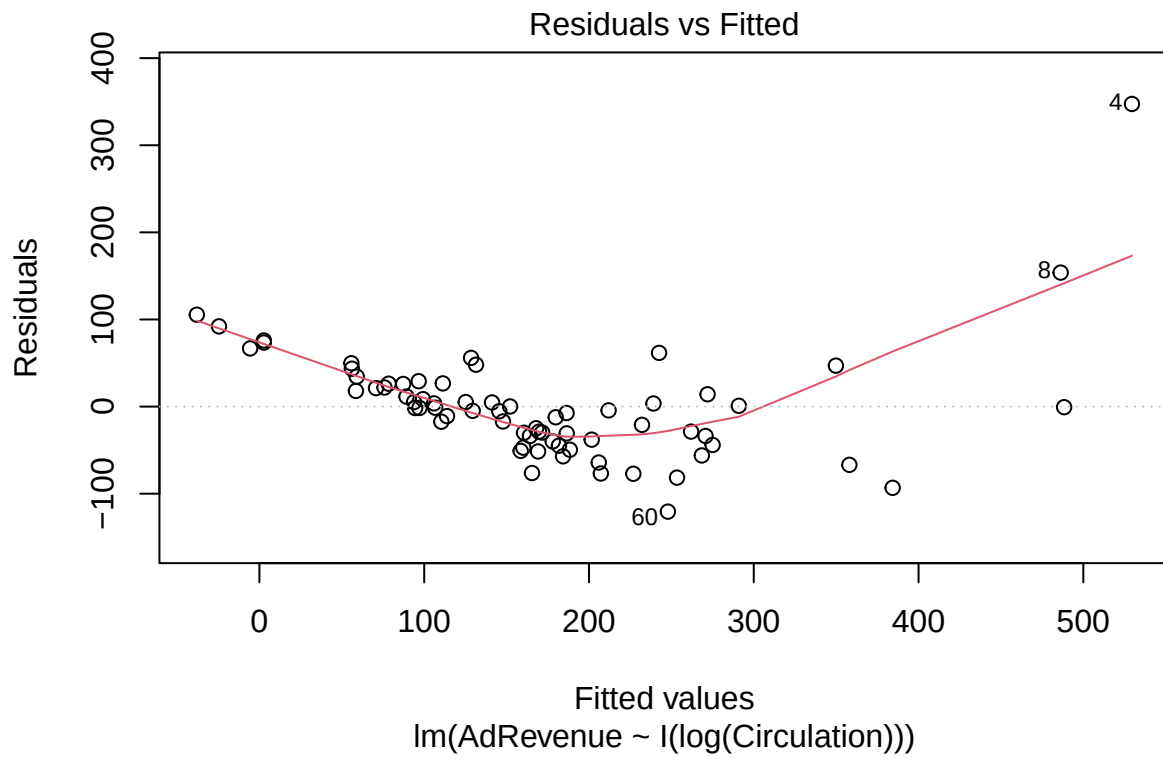


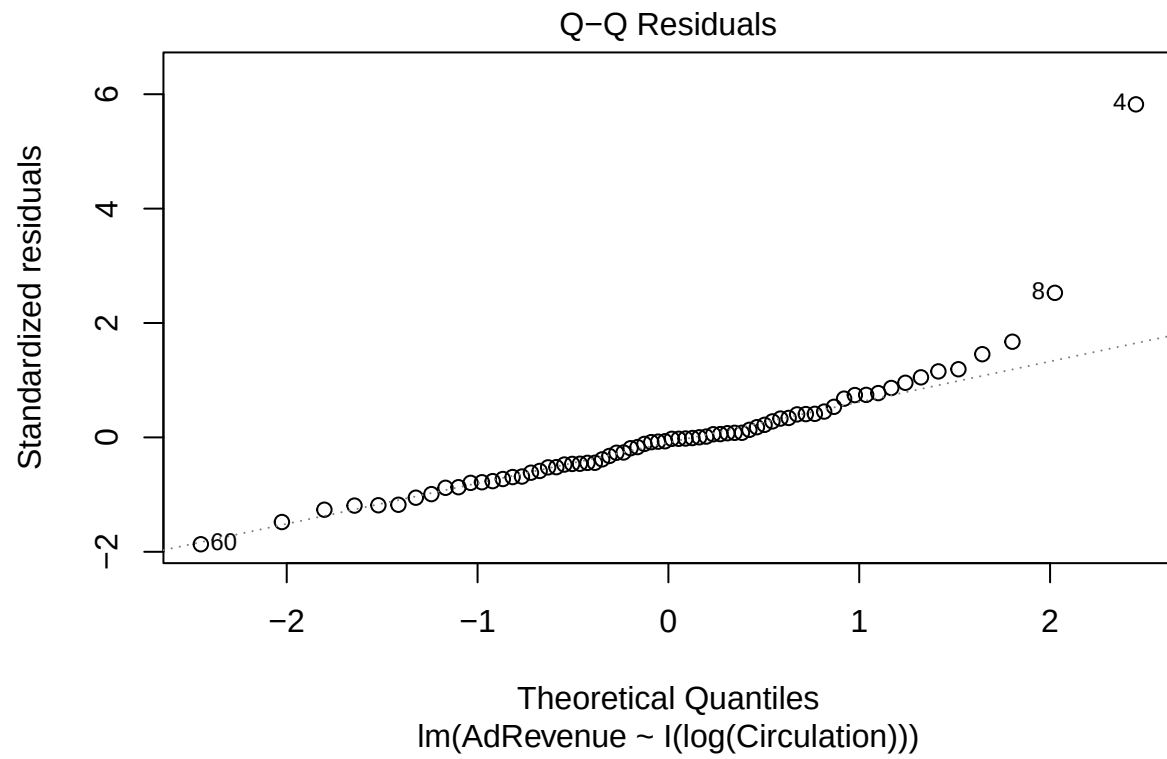
```
log_model <- lm(AdRevenue ~ I(log(Circulation)), df)
```

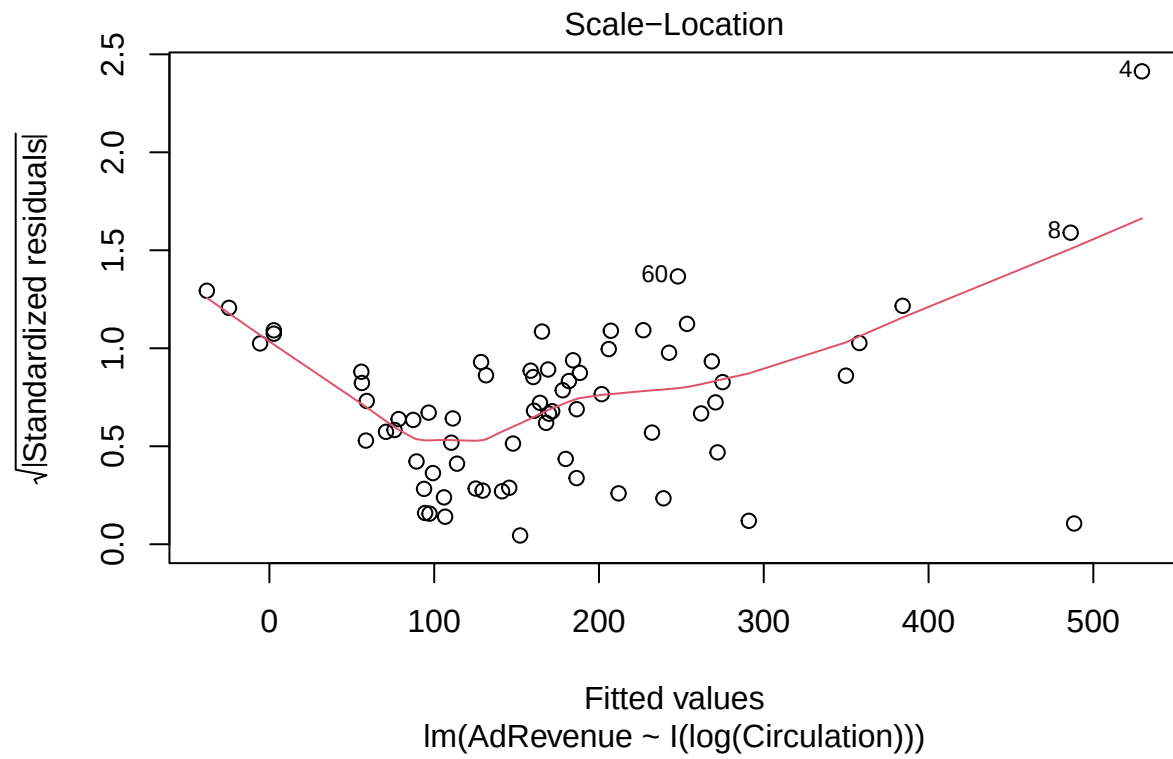
```
summary(log_model)
```

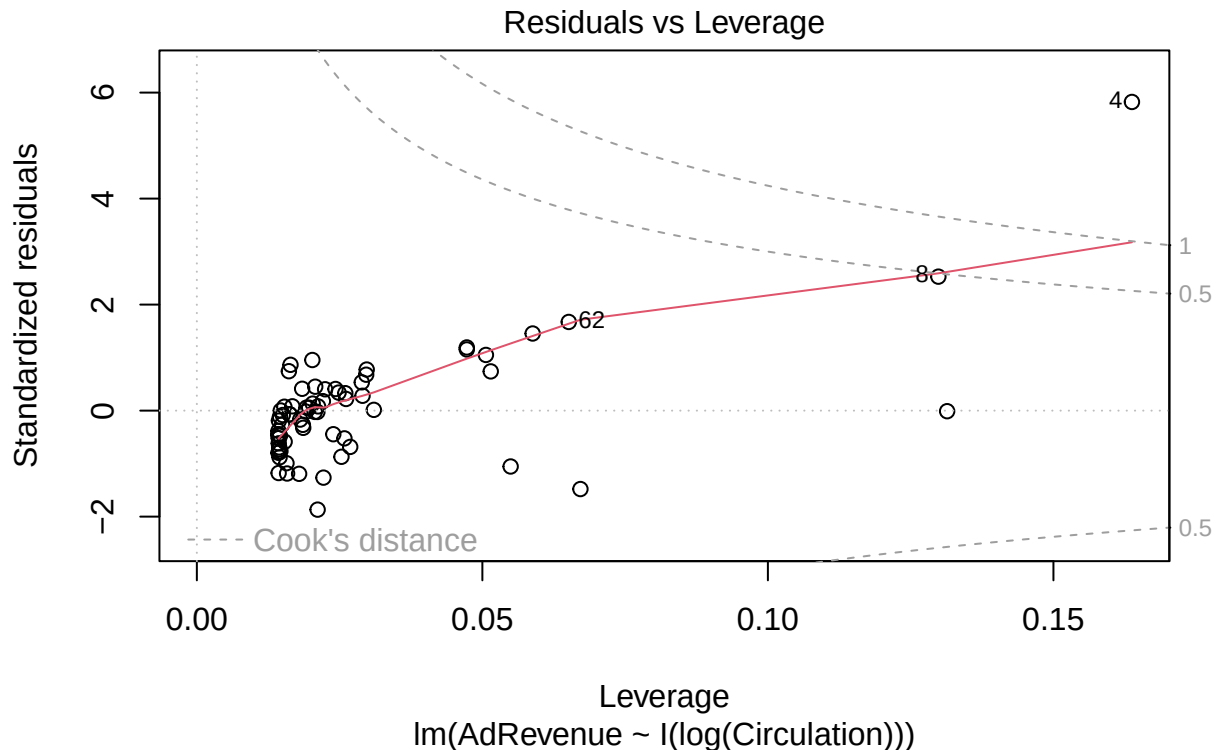
```
##
## Call:
## lm(formula = AdRevenue ~ I(log(Circulation)), data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -120.58  -36.92   -3.00   24.94   347.40
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    98.631     9.316   10.59 5.04e-16 ***
## I(log(Circulation)) 123.554     8.694   14.21 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 65.24 on 68 degrees of freedom
## Multiple R-squared:  0.7481, Adjusted R-squared:  0.7444
## F-statistic: 202 on 1 and 68 DF, p-value: < 2.2e-16
```

```
plot(log_model)
```







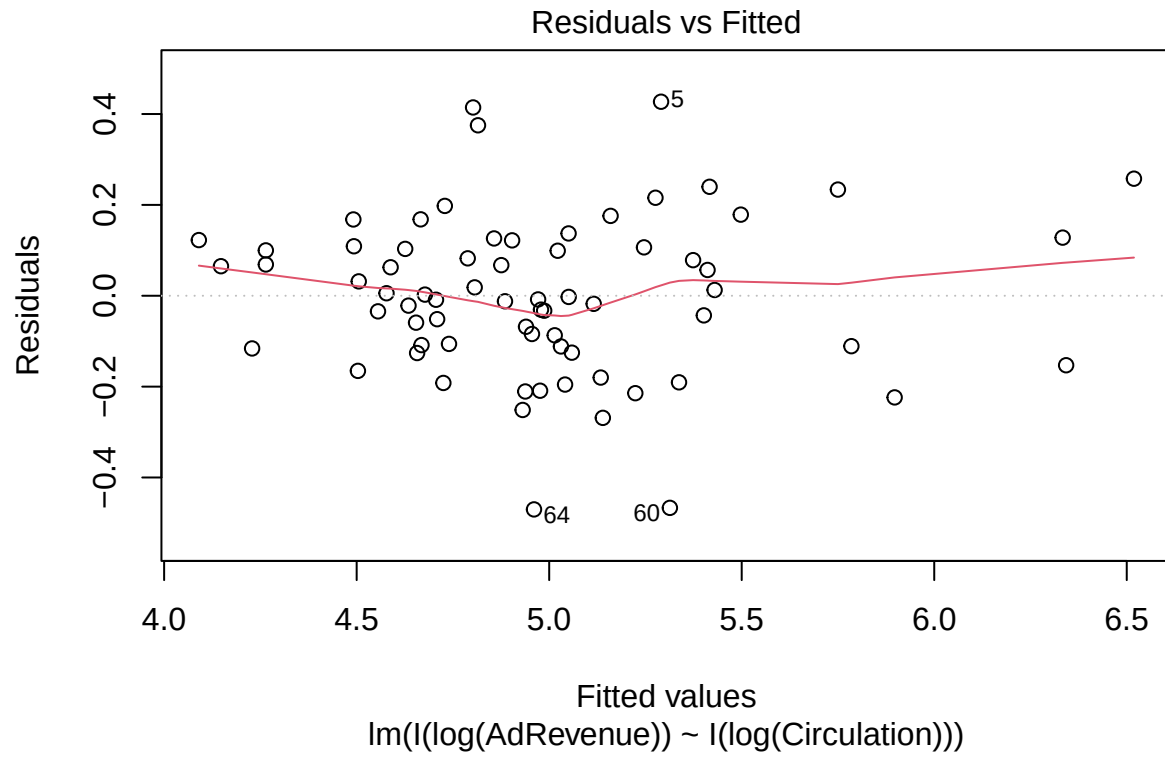


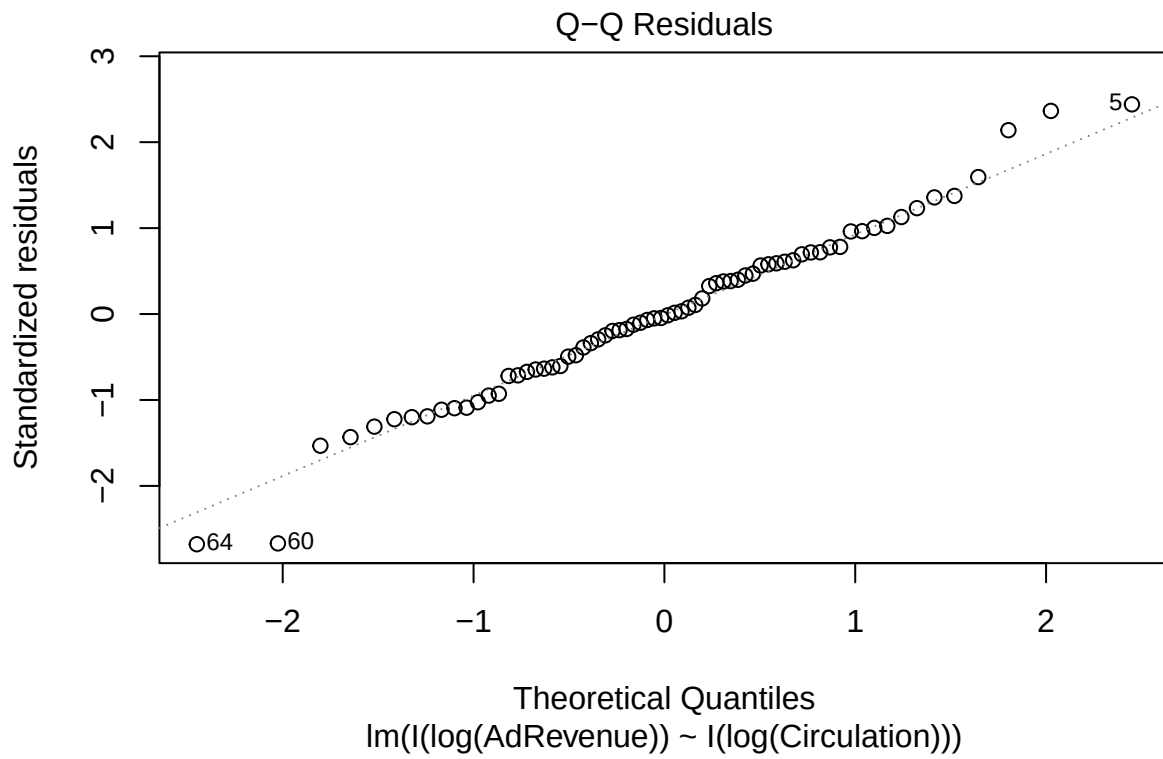
$\log(\text{Circulation})$ seems worse. Especially residual vs. fitted plot where there's an identifiable trend now. R-squared value is lower as well.

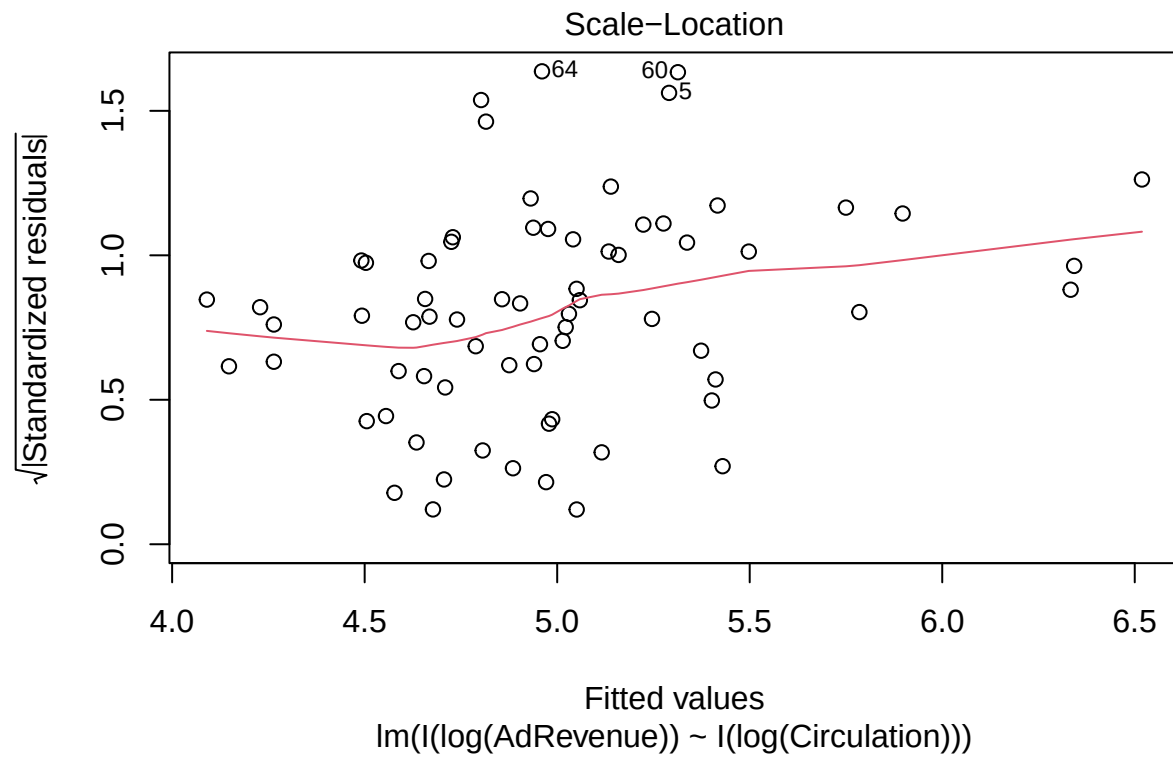
```
log_both_model <- lm(I(log(AdRevenue)) ~ I(log(Circulation)), df)
summary(log_both_model)
```

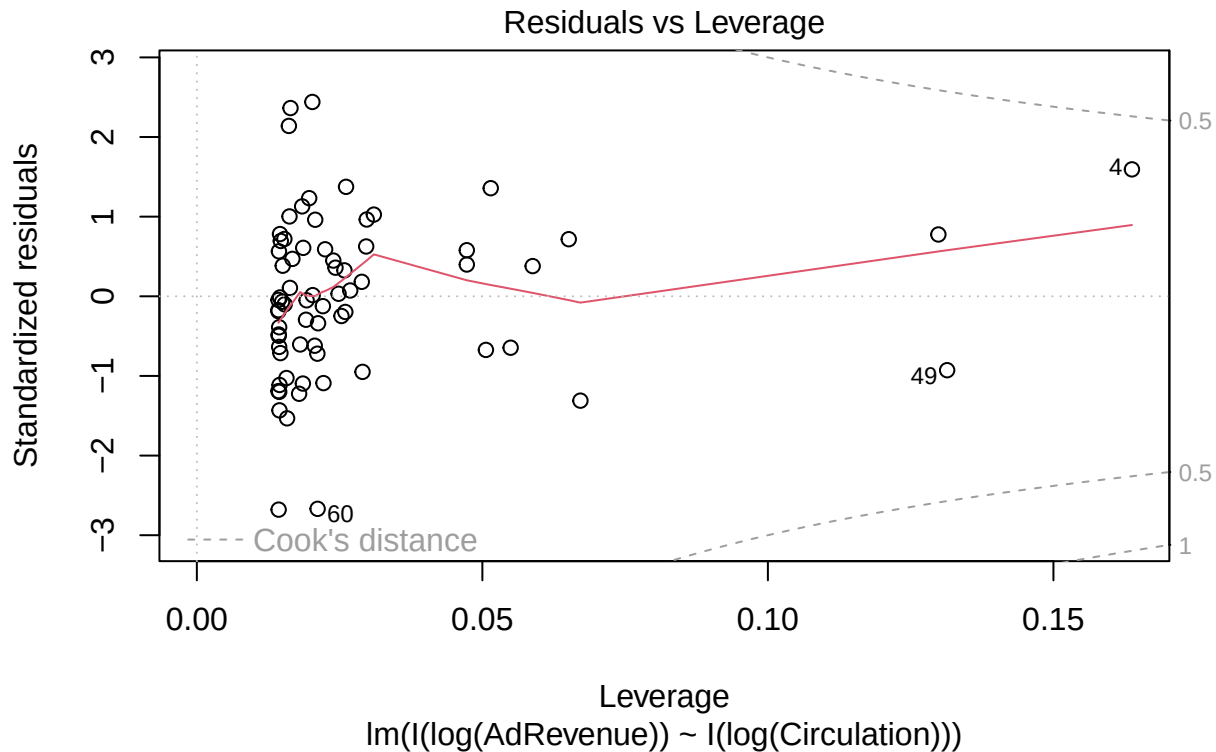
```
##
## Call:
## lm(formula = I(log(AdRevenue)) ~ I(log(Circulation)), data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.47022 -0.11142 -0.00532  0.10835  0.42705
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      4.67473    0.02525  185.16  <2e-16 ***
## I(log(Circulation)) 0.52876    0.02356   22.44  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1768 on 68 degrees of freedom
## Multiple R-squared:  0.881, Adjusted R-squared:  0.8793
## F-statistic: 503.6 on 1 and 68 DF, p-value: < 2.2e-16
```

```
plot(log_both_model)
```









This model is much better. Residuals vs. fitted is much better and there's a lack of a trend. QQ-plot seems much better as well where most values are in a straight line. Scale-location seems better. No bad high leverage either.

(b)

```
log_pred <- predict(log_both_model, newdata = data.frame(Circulation = c(0.5, 20)), interval = 'predict.
exp(log_pred)
```

```
##      fit      lwr      upr
## 1  74.30864  51.82406 106.5485
## 2 522.56626 359.89585 758.7626
```

74.30864 with (51.82406, 106.5485) 522.56626 with (59.89585 758.7626)

(c)

predictions become less intuitive, intervals are multiplicative and asymmetric. Constant variance of errors seem a bit iffy with the scale-location plot.

Part B

```

model1 <- lm(AdRevenue ~ Circulation, data = df)
model2 <- lm(AdRevenue ~ Circulation + I(Circulation^2), data = df)
model3 <- lm(AdRevenue ~ Circulation + I(Circulation^2) + I(Circulation^3), data = df)

```

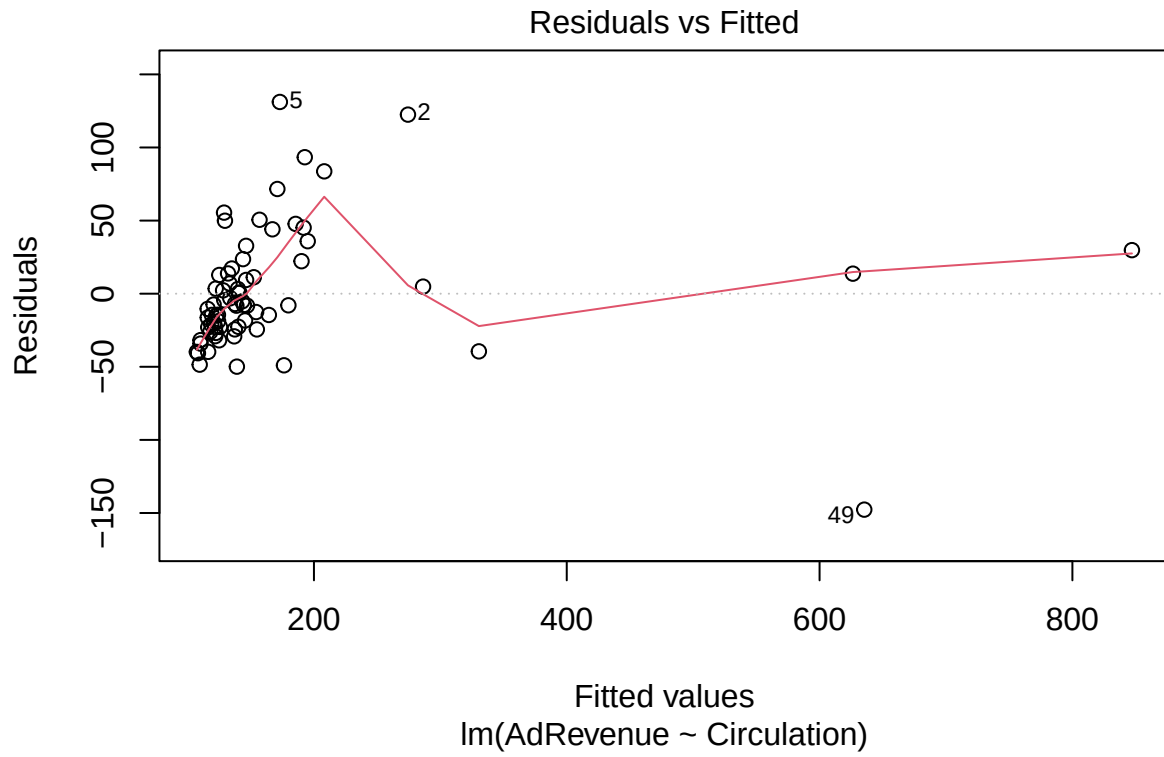
```
summary(model1)
```

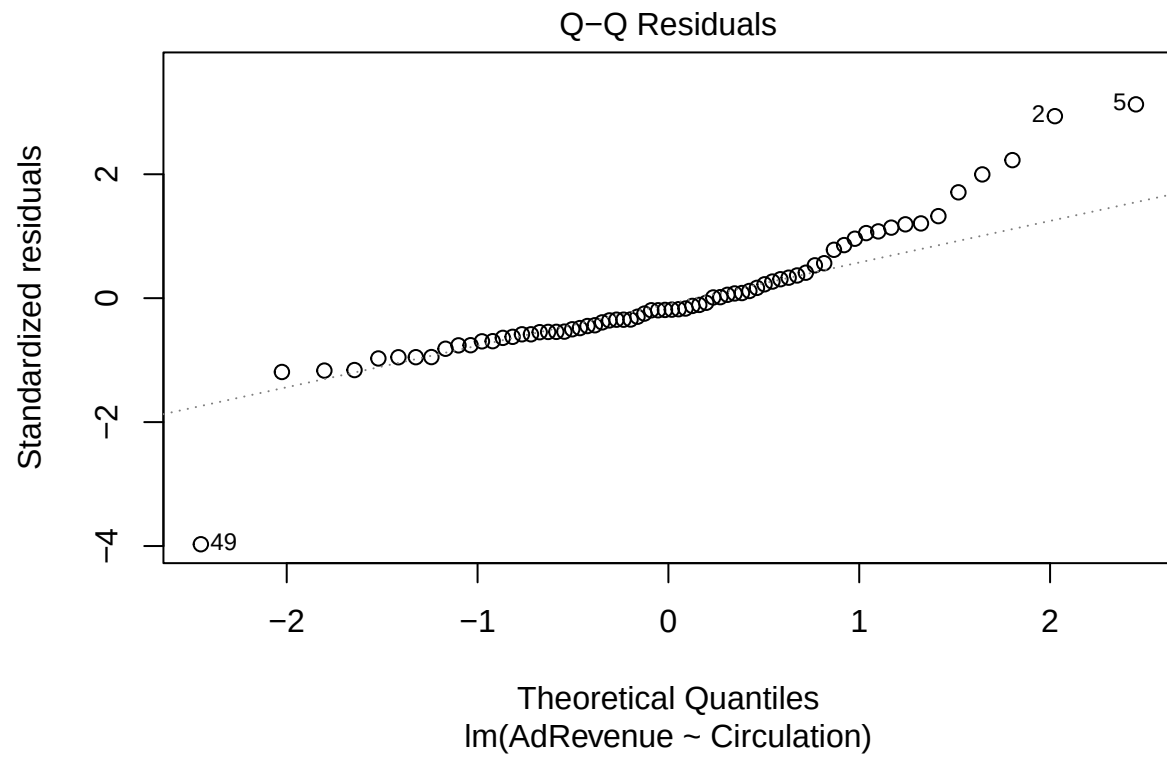
```

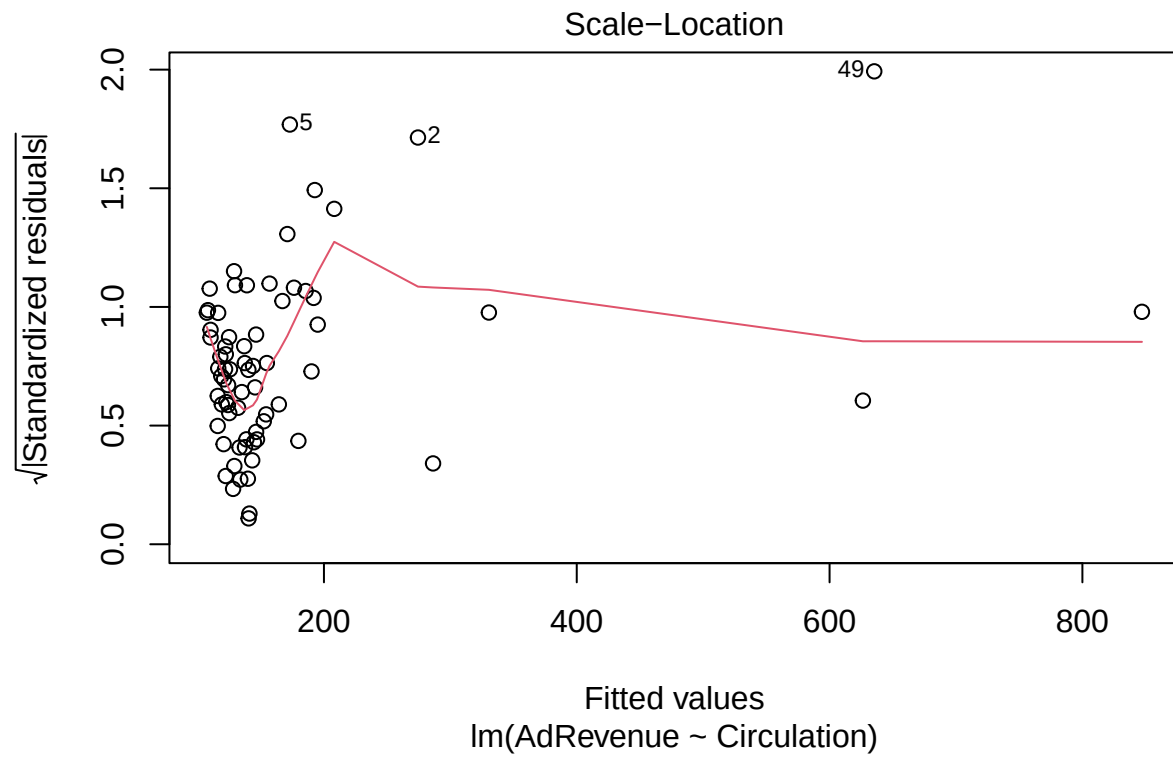
##
## Call:
## lm(formula = AdRevenue ~ Circulation, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -147.694  -22.939   -7.845   13.810  131.130
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  99.8095     5.8547   17.05  <2e-16 ***
## Circulation  22.8534     0.9518   24.01  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 42.22 on 68 degrees of freedom
## Multiple R-squared:  0.8945, Adjusted R-squared:  0.8929
## F-statistic: 576.5 on 1 and 68 DF,  p-value: < 2.2e-16

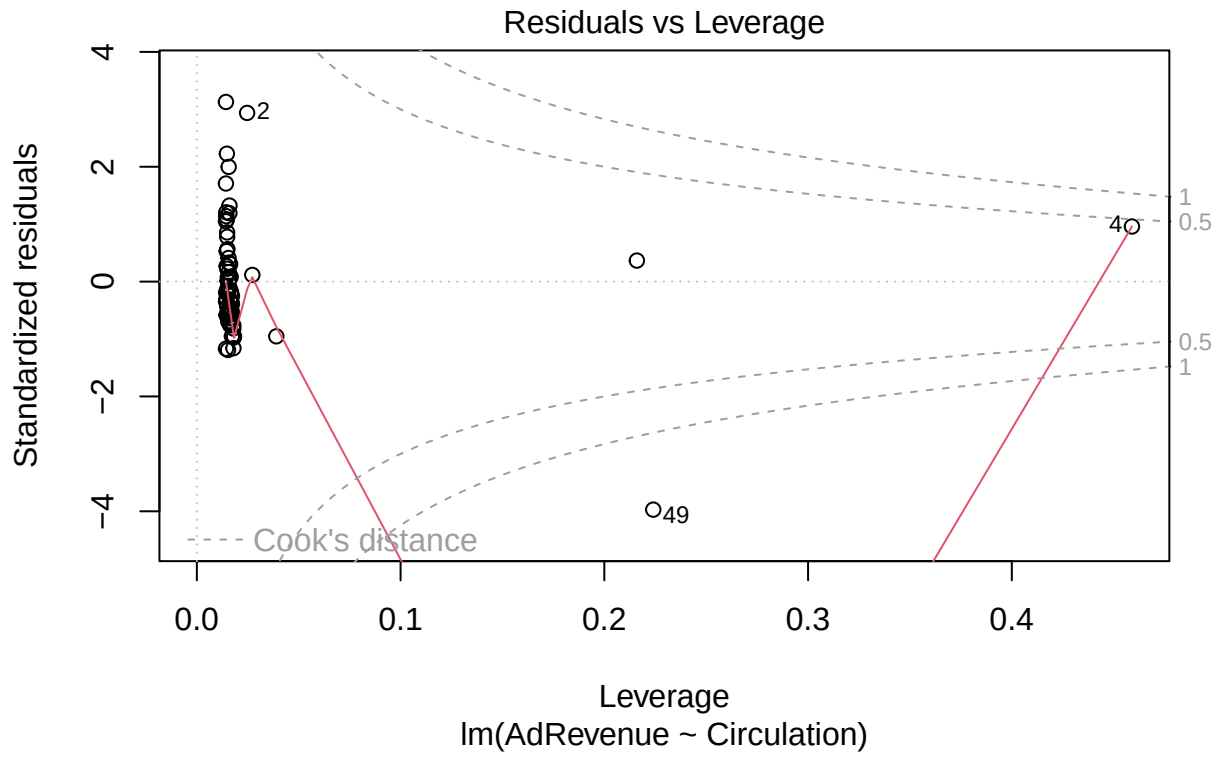
```

```
plot(model1)
```





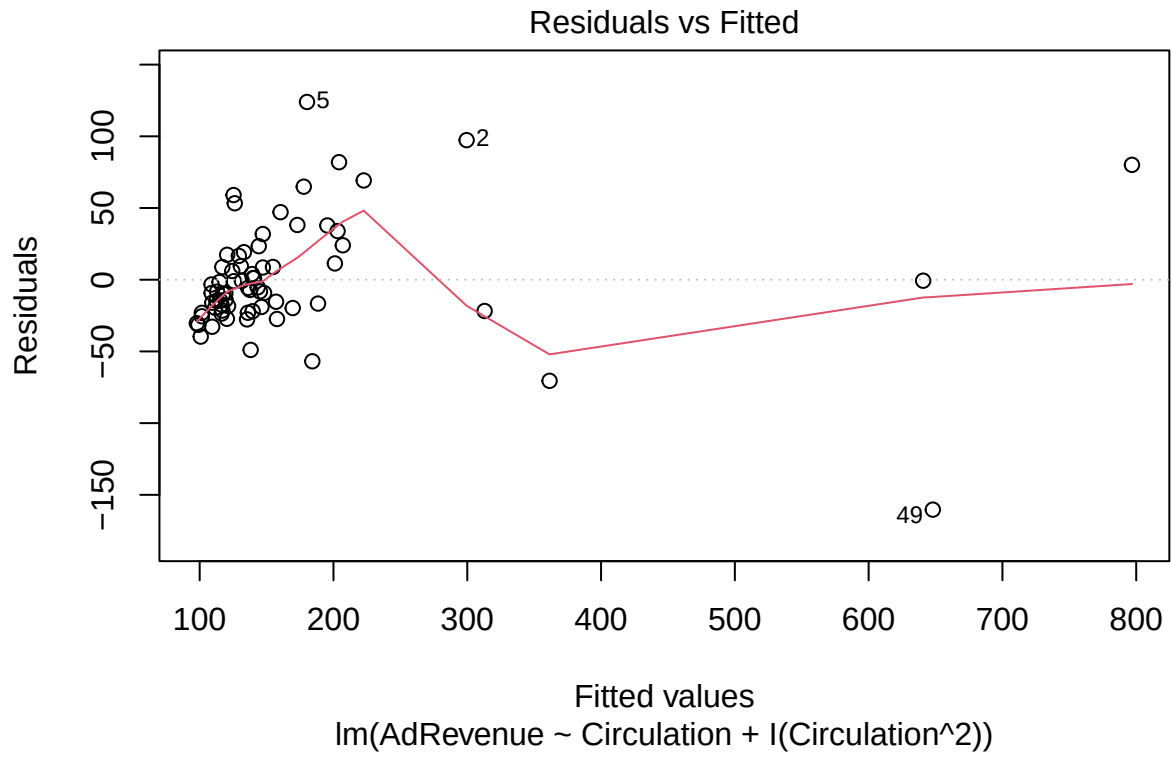


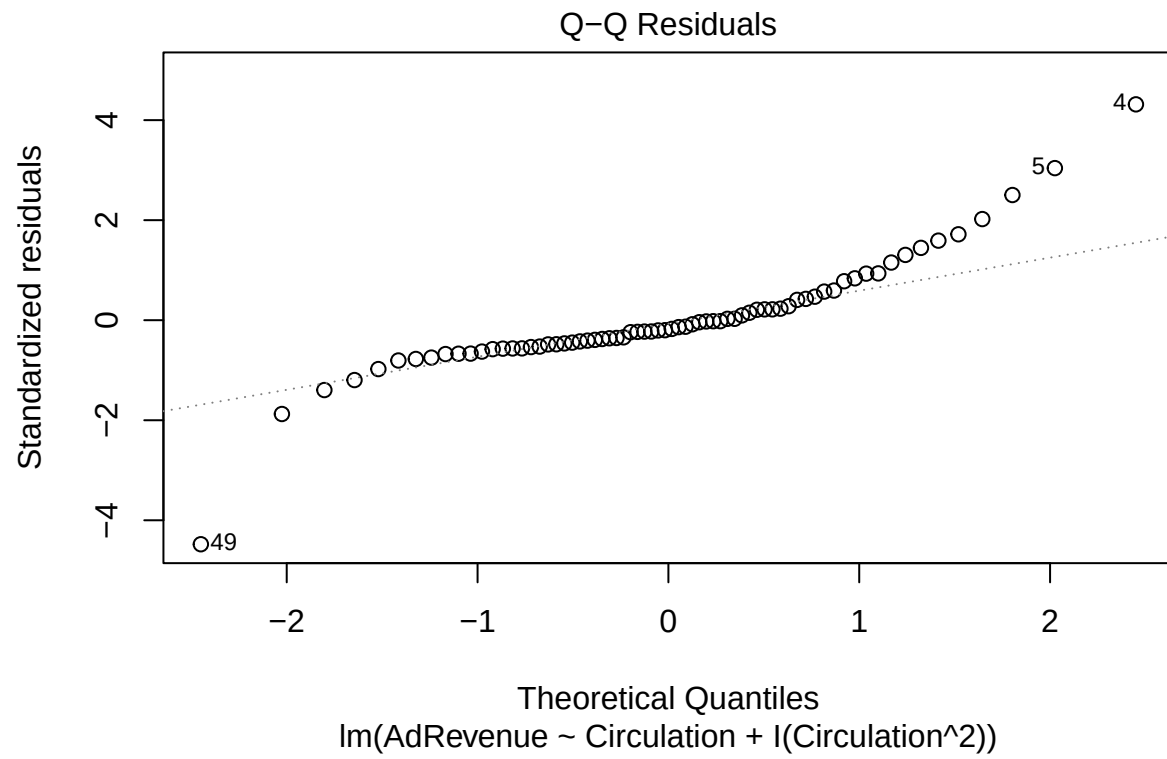


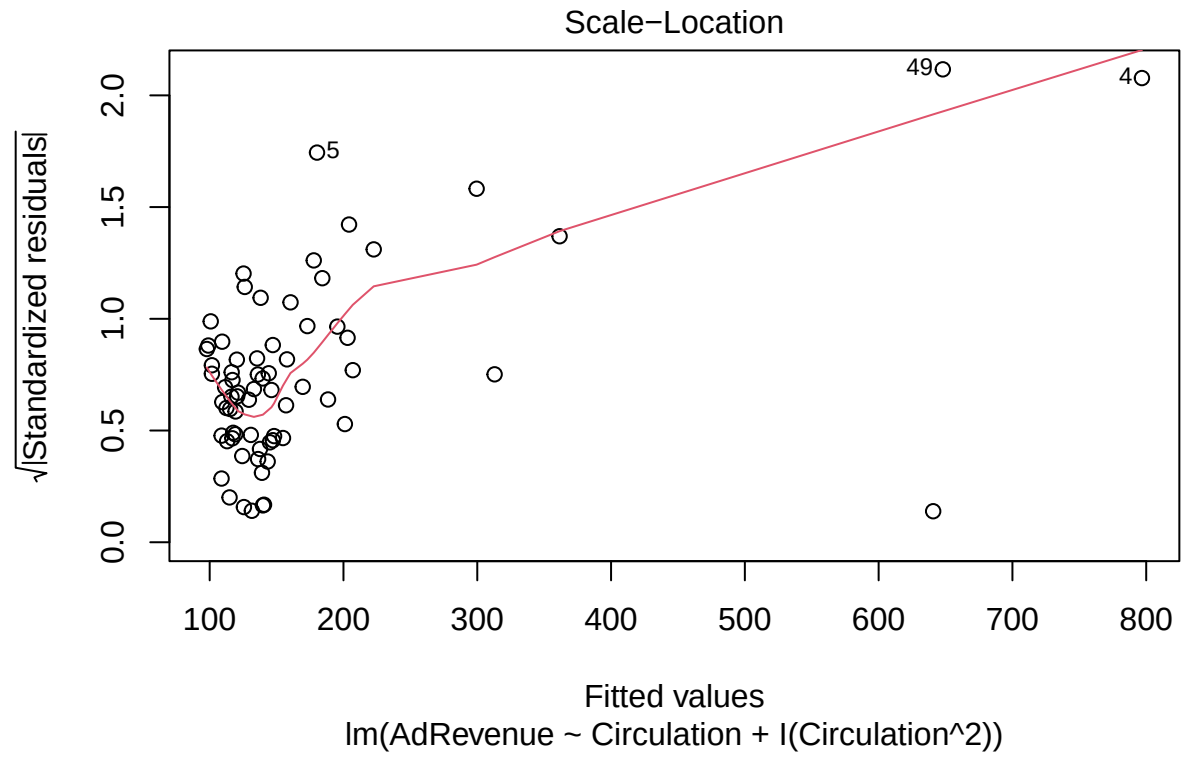
```
summary(model2)
```

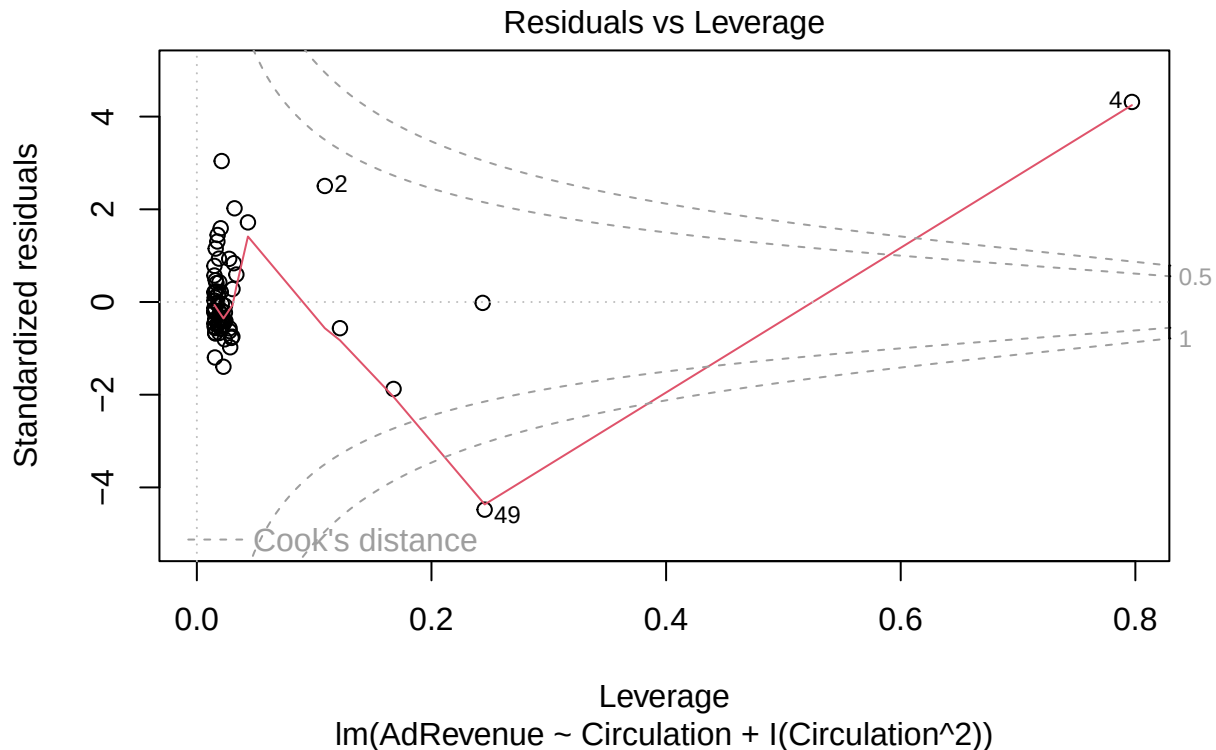
```
##
## Call:
## lm(formula = AdRevenue ~ Circulation + I(Circulation^2), data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -160.32  -21.05   -7.65   15.30  123.96
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    88.1390     7.9708  11.058 < 2e-16 ***
## Circulation     29.5006     3.2992   8.942 4.87e-13 ***
## I(Circulation^2) -0.2394     0.1140  -2.100  0.0395 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 41.2 on 67 degrees of freedom
## Multiple R-squared:  0.901, Adjusted R-squared:  0.8981
## F-statistic: 304.9 on 2 and 67 DF, p-value: < 2.2e-16
```

```
plot(model2)
```







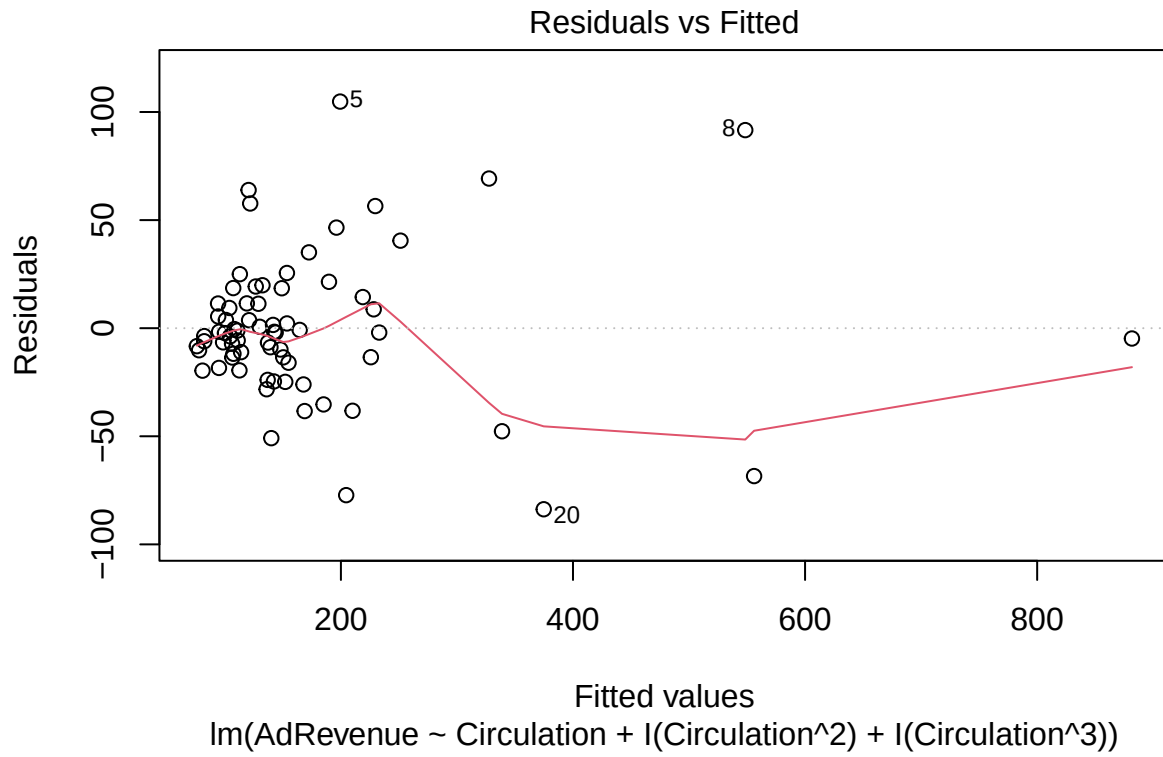


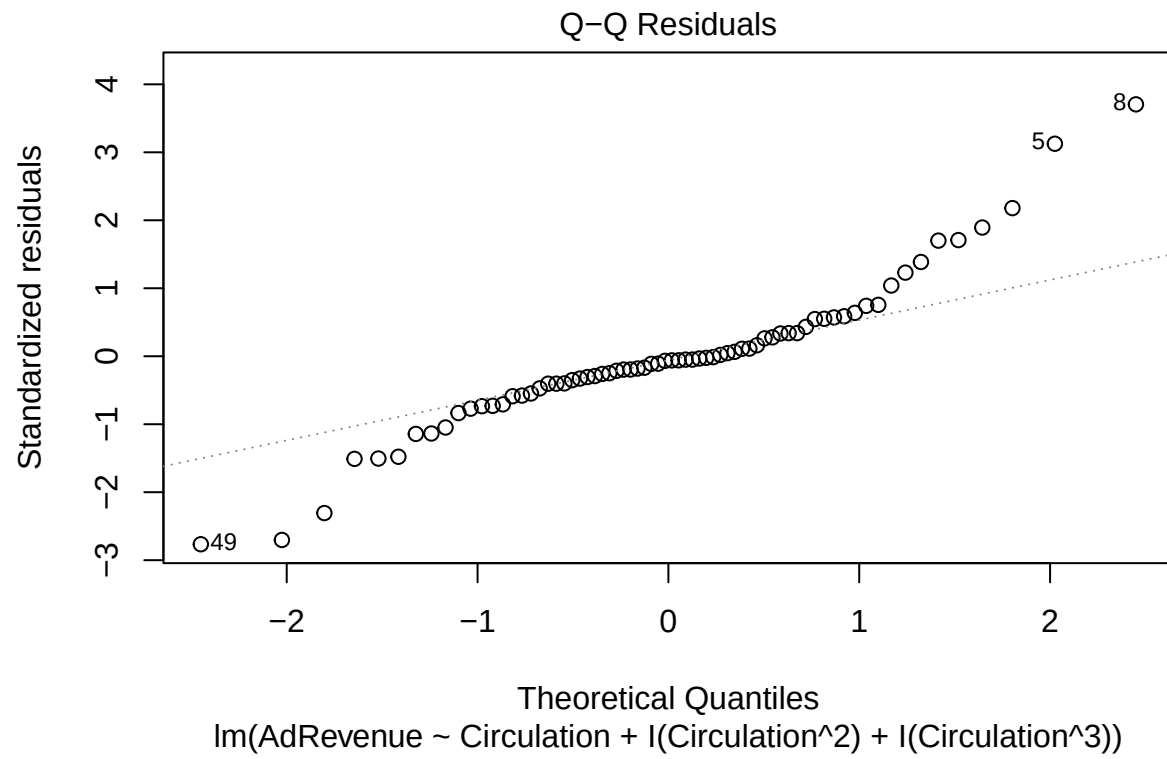
This one is no good. The upward-trend seen in the Scale-Location plot indicates that it is not valid in terms of the constant variance of errors. The QQ-plot is worse as well.

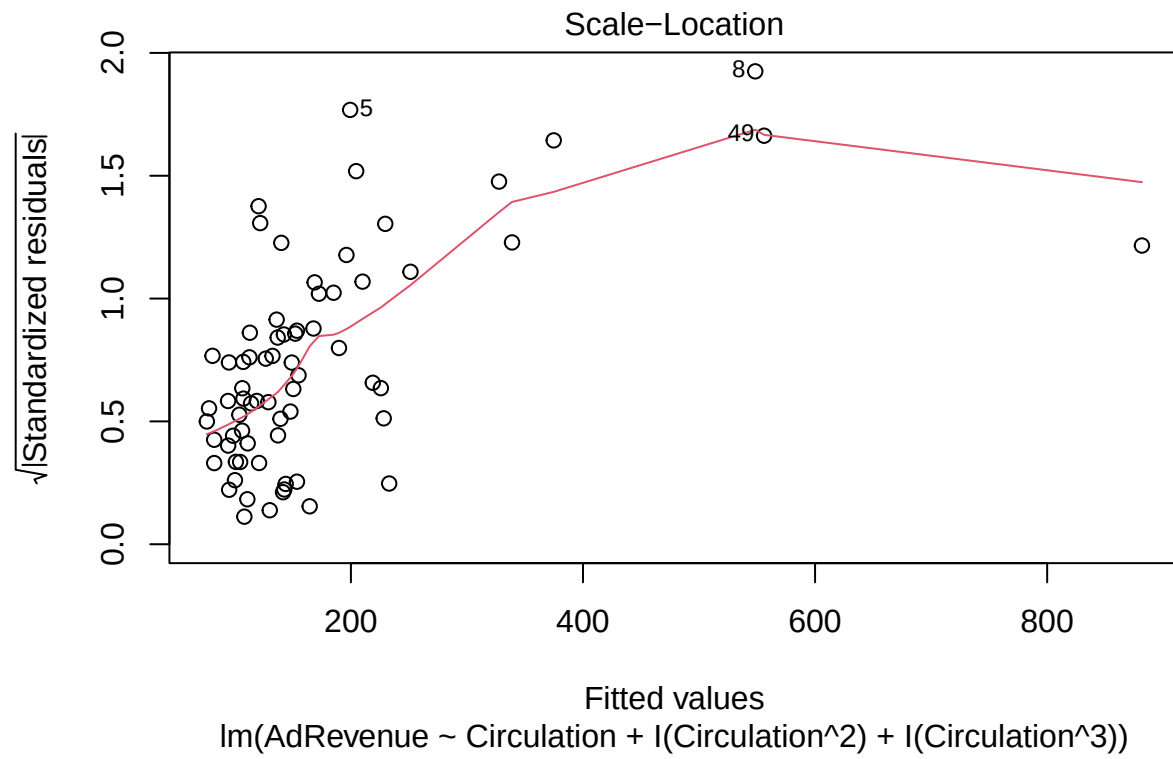
```
summary(model3)
```

```
##
## Call:
## lm(formula = AdRevenue ~ Circulation + I(Circulation^2) + I(Circulation^3),
##     data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -83.75 -13.56  -2.16   11.46  104.82
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    59.17037     8.34505   7.090 1.12e-09 ***
## Circulation    51.23582     4.71123  10.875 2.33e-16 ***
## I(Circulation^2) -2.50538     0.41141  -6.090 6.48e-08 ***
## I(Circulation^3)  0.05223     0.00923   5.658 3.57e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 34.06 on 66 degrees of freedom
## Multiple R-squared:  0.9333, Adjusted R-squared:  0.9303
## F-statistic: 308.1 on 3 and 66 DF,  p-value: < 2.2e-16
```

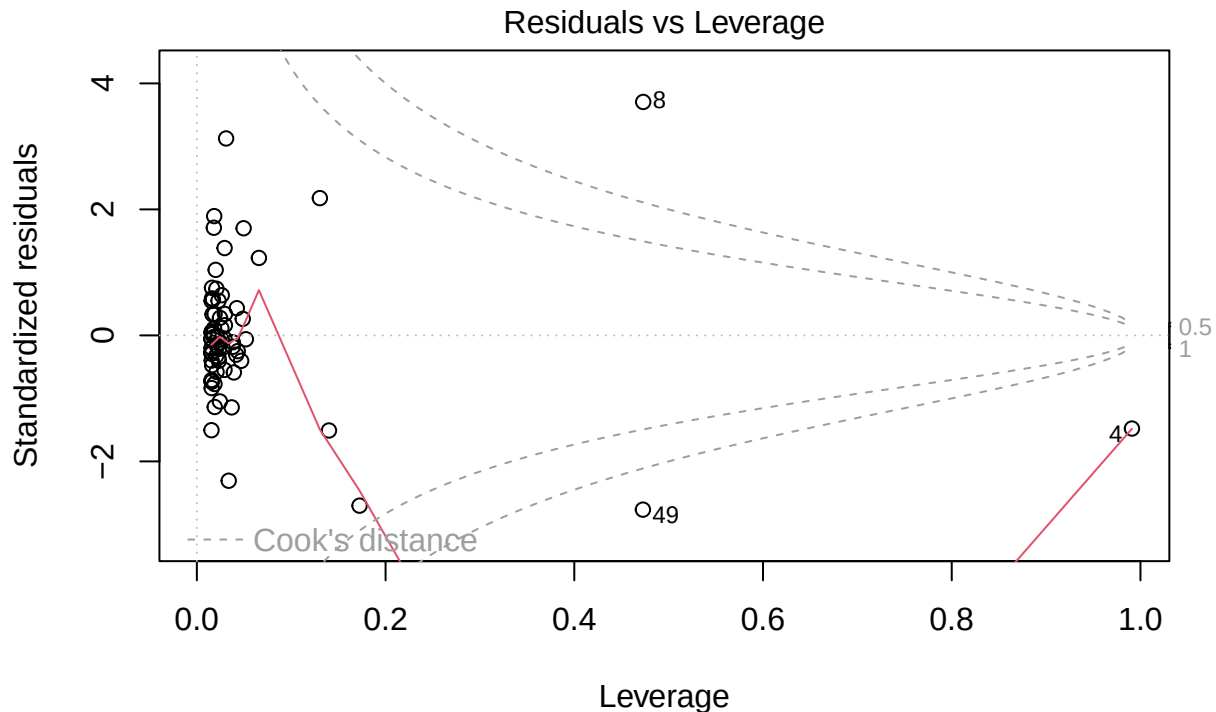
```
plot(model13)
```







```
## Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced
## Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced
```



$\text{lm}(\text{AdRevenue} \sim \text{Circulation} + \text{I}(\text{Circulation}^2) + \text{I}(\text{Circulation}^3))$

Scale-location is off again. QQ-plot is bad as well. Residuals plot ain't good either.

So the best choice of model is still the simple linear model without any polynomial transformations.

(b)

```
predict(model1, newdata = data.frame(Circulation = c(0.5, 20)), interval = 'prediction', level = 0.95)
```

```
##      fit      lwr      upr
## 1 111.2362  26.2383 196.2342
## 2 556.8774 466.1694 647.5854
```

111.2362 with (26.2383, 196.2342) 556.8774 with (466.1694, 647.5854)

(c)

There are outliers in the simple regression model as well as a residual plot that is not the best (all the values are congregated very closely at low fitted values)

Part C

(a)

Since the model in Part B is just the normal simple regression model, and since we compared that directly with our chosen model in Part A, then the model in Part A is better (double log transform), with reasons explained above. Examined the validity plots for the most part.

(b)

Part A: 74.30864 with (51.82406, 106.5485) 522.56626 with (59.89585 758.7626)

Part B: 111.2362 with (26.2383, 196.2342) 556.8774 with (466.1694, 647.5854)

I would recommend the bottom one (Part B) just because it's more intuitive and easy to interpret. Since the intervals in Part A were logged and then exponentiated, it will be harder to interpret.

(2)

```
df <- data.frame(read.csv('./marioKart.csv'))

str(df)

## 'data.frame':   141 obs. of  12 variables:
## $ ID          : num  1.5e+11 2.6e+11 3.2e+11 2.8e+11 1.7e+11 ...
## $ duration    : int   3 7 3 3 1 3 1 1 3 7 ...
## $ nBids       : int   20 13 16 18 20 19 13 15 29 8 ...
## $ cond        : chr   "new" "used" "new" "new" ...
## $ startPr     : num   0.99 0.99 0.99 0.99 0.01 ...
## $ shipPr      : num    4 3.99 3.5 0 0 4 0 2.99 4 4 ...
## $ totalPr     : num   51.5 37 45.5 44 71 ...
## $ shipSp      : chr   "standard" "firstClass" "firstClass" "standard" ...
## $ sellerRate  : int   1580 365 998 7 820 270144 7284 4858 27 201 ...
## $ stockPhoto  : chr   "yes" "yes" "no" "yes" ...
## $ wheels      : int    1 1 1 1 2 0 0 2 1 1 ...
## $ notes       : logi  NA NA NA NA NA NA NA ...
```

(a)

```
model <- lm(totalPr ~ startPr + shipPr + duration + nBids + sellerRate + wheels, data = df)

summary(model)

##
## Call:
## lm(formula = totalPr ~ startPr + shipPr + duration + nBids +
##     sellerRate + wheels, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6381 -3.0472  0.2475  2.1064 16.0556
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.483e+01  1.799e+00  19.362 < 2e-16 ***
## startPr      1.694e-01  3.782e-02   4.480 1.58e-05 ***
## shipPr       1.444e-01  1.590e-01   0.908  0.36529
## duration    -5.325e-01  1.811e-01  -2.940  0.00387 **
```

```
## nBids      2.375e-01  9.153e-02  2.595  0.01050 *
## sellerRate 2.308e-05  8.630e-06  2.674  0.00843 **
## wheels     7.938e+00  5.419e-01  14.649  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.736 on 134 degrees of freedom
## Multiple R-squared:  0.7415, Adjusted R-squared:  0.73
## F-statistic: 64.07 on 6 and 134 DF,  p-value: < 2.2e-16
```

Holding the rest of the predictors constant, a 1-unit increase in the number of bids is associated with an average change of +0.2375 dollars in the total price.

(b)

Holding all other variables constant, one more wheel is associated with an average change of +7.938 dollars in the total price.

(c)

```
model2 <- lm(totalPr ~ startPr + shipPr + duration + nBids + sellerRate + wheels + cond, data = df)
summary(model2)
```

```
##
## Call:
## lm(formula = totalPr ~ startPr + shipPr + duration + nBids +
##     sellerRate + wheels + cond, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.363 -2.527 -0.379  2.119 15.163
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.839e+01  1.827e+00  21.014  < 2e-16 ***
## startPr      1.533e-01  3.523e-02   4.352  2.66e-05 ***
## shipPr       1.256e-01  1.475e-01   0.851   0.3960
## duration    -2.286e-01  1.796e-01  -1.273   0.2052
## nBids        1.754e-01  8.586e-02   2.043   0.0431 *
## sellerRate   1.762e-05  8.084e-06   2.180   0.0310 *
## wheels       7.077e+00  5.338e-01  13.257  < 2e-16 ***
## condused    -4.455e+00  9.323e-01  -4.778  4.61e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.392 on 133 degrees of freedom
## Multiple R-squared:  0.7794, Adjusted R-squared:  0.7678
## F-statistic: 67.13 on 7 and 133 DF,  p-value: < 2.2e-16
```

Holding all variables constant, a used game is associated with an average change of -4.45 dollars in total price compared to a new game. Yes, it does matter with all things being equal – it is worth 4.45 dollars less on average.

(3)

```
df <- data.frame(read.csv('./practiceproblem.csv'))

str(df)

## 'data.frame':    7481 obs. of  3 variables:
## $ finish_time : int  433 410 471 385 554 421 439 471 469 406 ...
## $ barrier_time: int  123 119 131 110 157 120 128 137 135 116 ...
## $ resids      : num  -5.32 -14.31 4.66 -7.78 -3.42 ...

y <- df$finish_time
x <- df$barrier_time
resids <- df$resids
```

(a)

```
# SXX = variation of x
SXX <- sum((x - mean(x))^2)
SXX
```

```
## [1] 892547.2
```

(b)

```
# SY Y = variation of y
SY Y <- sum((y - mean(y))^2)
SY Y
```

```
## [1] 12148286
```

(c)

```
# estimate =  $s^2 = SY Y / (n - 1)$ 

var <- SY Y / (length(df) - 1)

var
```

```
## [1] 6074143
```

(d)

```
##  $SSY = SSreg + RSS$ ;  $SSreg = SSY - RSS$ 
```

```
SSY <- SY
RSS <- sum(resids^2)

SSreg <- SSY - RSS

SSreg
```

```
## [1] 10951384
```

(e)

```
##  $R\text{-squared} = SSreg / SSY$ 
```

```
R_squared <- SSreg / SSY

R_squared
```

```
## [1] 0.9014756
```

(checking)

```
model <- lm(y ~ x)
```

```
summary(model)
```

```
##
## Call:
## lm(formula = y ~ x)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -226.960   -7.333   -1.310    6.198  142.198
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.47376    1.66978   4.476 7.72e-06 ***
## x            3.50283    0.01339 261.593 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.65 on 7479 degrees of freedom
## Multiple R-squared:  0.9015, Adjusted R-squared:  0.9015
## F-statistic: 6.843e+04 on 1 and 7479 DF, p-value: < 2.2e-16
```

```
anova(model)
```

```
## Analysis of Variance Table
##
## Response: y
##           Df    Sum Sq Mean Sq F value    Pr(>F)
## x             1 10951384 10951384   68431 < 2.2e-16 ***
## Residuals 7479  1196902      160
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```